

Electrochemical performance of LiFePO_4 cathode material coated with ZrO_2 nanolayer

H. Liu, G.X. Wang^{*}, D. Wexler, J.Z. Wang, H.K. Liu

*School of Mechanical, Materials and Mechatronics Engineering, Institute for Superconducting and Electronic Materials,
University of Wollongong, Wollongong, NSW 2522, Australia*

Received 29 October 2007; received in revised form 13 November 2007; accepted 14 November 2007
Available online 24 November 2007

Abstract

ZrO_2 nanolayer coated LiFePO_4 particles were successfully prepared by a chemical precipitation method. Nanolayer structured ZrO_2 was found on the surface of LiFePO_4 particles by high resolution transmission electron microscopy (HRTEM). The coating does not affect the crystal structure of the LiFePO_4 core, as determined by X-ray diffraction (XRD) and selected area electron diffraction (SAED) on individual particles. The ZrO_2 coating can remarkably improve the electrochemical performance at high charge/discharge rate. This improvement may be due to the amelioration of the electrochemical dynamics on the LiFePO_4 electrode/electrolyte interface resulting from the effects of the ZrO_2 nanolayer coating.

© 2007 Elsevier B.V. All rights reserved.

Keywords: Lithium ion batteries; Cathode material; LiFePO_4 ; Surface coating

1. Introduction

Since the first report in 1997, LiFePO_4 has been widely investigated as a promising cathode material to replace LiCoO_2 for lithium ion batteries. LiFePO_4 cathode material demonstrates reversible lithium intercalation at around 3.4 V (vs. Li^+/Li) with a theoretical capacity of 170 mAh/g [1,2]. LiFePO_4 is the most attractive cathode material for large-scale high power lithium ion batteries, due to its low cost, abundant raw materials and environmental benignity. The main obstacle to LiFePO_4 is its low electronic and ionic conductivity. At room temperature, the electronic conductivity of pristine LiFePO_4 is only 10^{-9} – 10^{-10} S/cm [3], which is much lower than those of LiCoO_2 ($\sim 10^{-3}$ S/cm) and LiMn_2O_4 (2×10^{-5} – 5×10^{-5} S/cm) [4,5]. In order to overcome this drawback, conductive agent coating is used to enhance the electronic conductivity. Carbon coating is a common method to overcome the limited rate capacity, because the dispersed carbon pro-

vides pathways for electron transference, resulting in improvement of the conductivity and electrochemical properties [6–9]. In addition, small metal particles and conductive polymers are also used as coating materials [10,11]. On the other hand, it has been reported that doping by supervalent cations increases the electronic conductivity of LiFePO_4 by a factor of $\sim 10^8$ [3]. Other studies of alien cations doping, either on Li sites (M1) or Fe sites (M2), have indicated that the electrochemical performance of LiFePO_4 could be improved by doping [12–14]. However, there has been controversy about whether this doping mechanism actually exists [15,16]. Islam et al. did some theoretical research on this and concluded that LiFePO_4 is not tolerant to aliovalent doping on either Li (M1) or Fe (M2) sites [16]. Therefore, there is no improving effect on the electrochemical performance of LiFePO_4 through aliovalent doping.

According to the previous reports, metal oxide coating is favorable to improving the electrochemical properties of cathode materials such as LiCoO_2 , LiNiO_2 and LiMn_2O_4 [17–21]. ZrO_2 , Al_2O_3 , TiO_2 and B_2O_3 are used as coatings on LiCoO_2 to make a zero-strain cathode by restraining

^{*} Corresponding author. Fax: +61 2 42213700.

E-mail address: gwang@uow.edu.au (G.X. Wang).

the H-M-H transformation during the charge/discharge process. The cycling performance of coated LiCoO_2 increases in accordance with the following order: $\text{B}_2\text{O}_3 < \text{TiO}_2 < \text{Al}_2\text{O}_3 < \text{ZrO}_2$, which is consistent with their toughness [18]. However, there has been no report as yet on oxide coating on the LiFePO_4 particle surface. Herein, we report for the first time on ZrO_2 nanocoating on the LiFePO_4 particle surface and on the resultant electrochemical performance enhancement. The electrochemical properties of the coated LiFePO_4 as cathode material were examined by cyclic voltammetry and constant charge/discharge testing.

2. Experimental

The ZrO_2 coating was carried out by a chemical precipitation method, using the commercial LiFePO_4 from DLG Battery Co. Ltd, China. The as-received LiFePO_4 materials contain less than 1% carbon as determined by the TGA analysis. The LiFePO_4 was mixed in distilled water with magnetic stirring. Then, one or two drops of dilute sulfuric acid was added to create activated sites on the LiFePO_4 surface. A 0.1 M ZrCl_4 solution was added in and stirred for 2 h. $\text{Zr}(\text{OH})_4$ precipitation was induced by adding the sodium hydroxide until the pH value was between 7 and 8. The mixture was filtered and washed with distilled water, then dried at 50 °C overnight. Finally, the powder was sintered at 600 °C for 2 h in inert atmosphere. The nominal ZrO_2 percentage was maintained at 1 wt%.

The cathode electrodes were fabricated by mixing the active materials with acetylene black (AB) and a binder poly(vinylidene fluoride) (PVdF) at a weight ratio of 85:10:5, respectively, in *N*-methyl-2-pyrrolidone (NMP) solvent. The slurry was uniformly pasted on Al foil with a blade. CR2032 coin cells were assembled for electrochemical testing. The electrolyte was 1 M LiPF_6 in a 1:1 mixture of ethylene carbonate and dimethyl carbonate.

The microstructure and morphology of the bare and coated LiFePO_4 were characterized by X-ray diffraction (XRD, Philips 1730), scanning electron microscopy (SEM, JEOL JEM-3000), and transmission electron microscopy (TEM, JEOL 2011). The cells were galvanostatically charged and discharged at a current density of 0.1 C or 1 C within a range of 2.5–4.2 V. The cyclic voltammetry (CV) curves were measured at 0.1 mV/s within the range of 2.5–4.5 V. In the electrochemical impedance spectroscopy (EIS) measurements, the excitation voltage applied to the cells was 5 mV and the frequency range was between 100 kHz and 10 mHz.

3. Results and discussion

Fig. 1 shows the XRD patterns of bare and ZrO_2 coated LiFePO_4 powders. All diffraction lines can be indexed to the ordered orthorhombic olivine crystal structure (space group: P_{nmb}). However, no diffraction lines corresponding to monoclinic ZrO_2 phase appeared in the XRD pattern

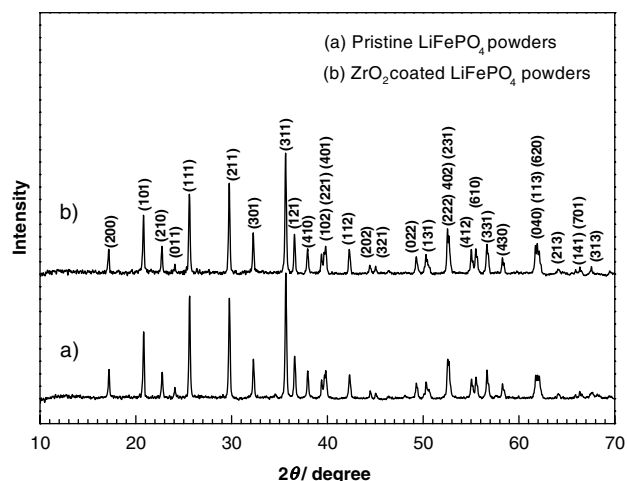


Fig. 1. X-ray diffraction patterns of (a) bare LiFePO_4 and (b) ZrO_2 coated LiFePO_4 .

of the coated LiFePO_4 . This suggests the formation of a thin-film coating layer on the surface of the LiFePO_4 particles. We performed SEM observations on the bare and ZrO_2 coated LiFePO_4 particles. In general, the particle size distribution becomes more homogeneous after the coating treatment (not shown here). Energy dispersive X-ray (EDX) analysis and elemental mapping detected the presence of the element Zr and homogeneous distribution of ZrO_2 on the surface of the LiFePO_4 particles, implying that the core particles were fully coated by the ZrO_2 layer.

The microstructures of the ZrO_2 coated LiFePO_4 particles were further analysed by TEM and high resolution TEM analysis. Fig. 2a shows a TEM image of a single coated LiFePO_4 crystal, which has a smooth surface and a diameter of about 300 nm. The selected area electron diffraction (SAED) pattern of this particle is presented in the inset of Fig. 2a. It shows no diffraction spots from ZrO_2 phase. The edge of a ZrO_2 coated LiFePO_4 particle is shown in Fig. 2b. This HRTEM image clearly illustrates the ZrO_2 coating layer. The thickness of the coating layer is about 2–3 nm. The interplanar spacing of 0.53 nm is consistent with the standard *d*-space of the (020) planes of orthorhombic LiFePO_4 . We performed spot EDX analysis at the coating layer, detecting the presence of Zr and O. This confirmed that the coating consists of ZrO_2 film.

Cyclic voltammogram (CV) profiles of the bare and ZrO_2 coated LiFePO_4 , collected at a scanning rate of 0.1 mV/s are shown in Fig. 3a. The well-defined reaction peaks of the bare LiFePO_4 at 3.660/3.275 V can be attributed to the $\text{Fe}^{3+}/\text{Fe}^{2+}$ redox couple transformation, accompanied by lithium ion insertion/extraction into and out of the electrode. The main $\text{Fe}^{3+}/\text{Fe}^{2+}$ redox peaks of ZrO_2 coated LiFePO_4 are located at 3.591/3.276 V. The decrease in the redox potential separation implies that the electrode reactivity is improved by the ZrO_2 coating. Fig. 3b shows the a.c. impedance spectra of the bare and coated LiFePO_4 electrode, which was measured in the fully discharged state after cycling. The depressed semicircle in

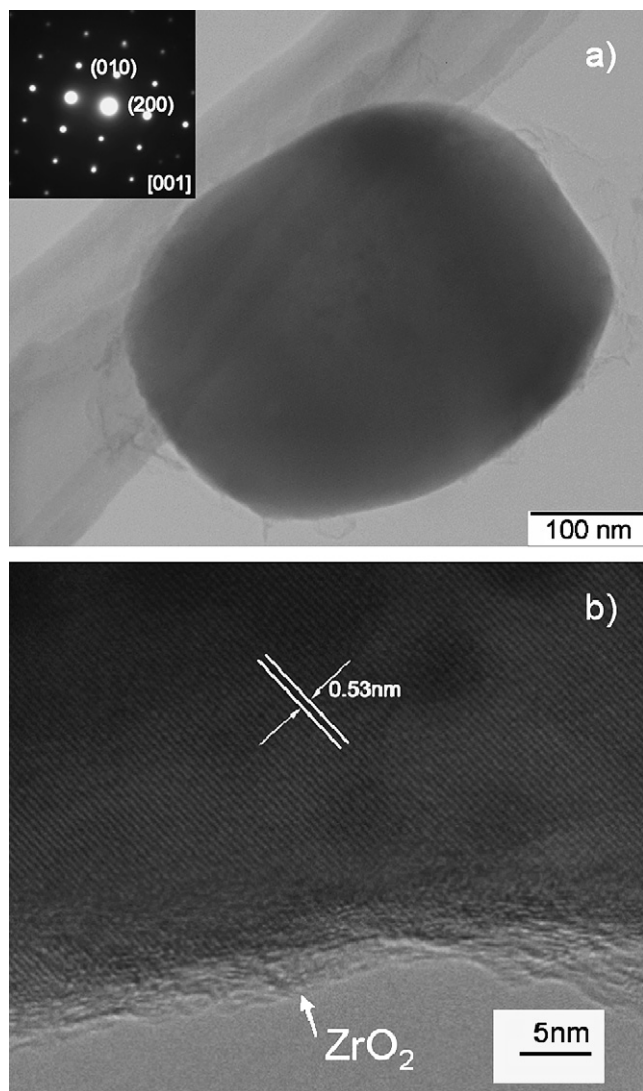


Fig. 2. (a) TEM image of a single ZrO_2 coated LiFePO_4 particle. The inset is the SAED pattern (b) HETEM image of the edge of a ZrO_2 coated LiFePO_4 particle.

the moderate frequency region is attributed to the charge transfer process. The numerical value of the diameter of the semicircle on the Z_{re} axis is approximate equal to the charge transfer resistance (R_{ct}) or the so-called the interfacial resistance. Comparing the semicircles of the bare and coated samples in the moderate frequency region, it is evident that the R_{ct} was decreased by the ZrO_2 coating. The improved particle surface due to the ZrO_2 nanolayer coating probably is more favorable for lithium ions migration, and therefore enhances the electrochemical performance. This scenario is similar to that proposed for $\text{Li}_4\text{Ti}_5\text{O}_{12}$ coating on LiCoO_2 cathode, which reduced the interfacial resistance, an effect that is mainly attributed to the suppression of the development of a space charge layer on the surface of the electrode [22].

The charge/discharge profiles at 0.1 C are shown in Fig. 4a. We noted that the voltage plateau is lengthened for the ZrO_2 coated electrode, which should be attributed

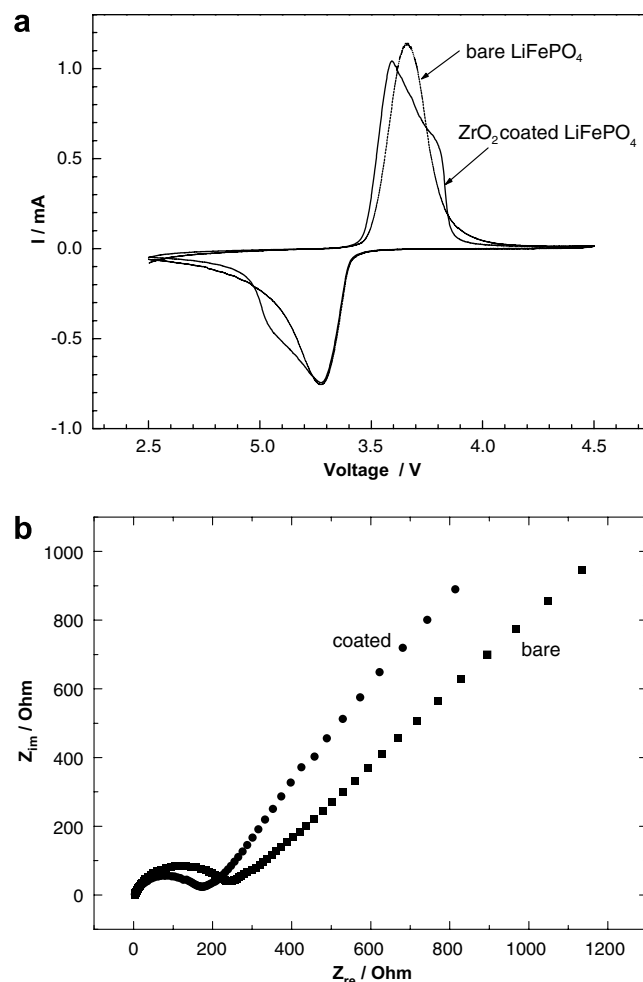


Fig. 3. (a) CV profiles of the bare and ZrO_2 coated LiFePO_4 at the scanning rate 0.1 mV/s. (b) EIS spectra of the bare and ZrO_2 coated LiFePO_4 electrode in the frequency range between 100 kHz and 10 mHz.

to the lower degree of polarization of the electrode materials during the electrochemical reaction after coating. The cyclabilities of the bare and coated LiFePO_4 at the 0.1 C and 1 C rates are shown in Fig. 4b. The capacities of both electrodes gradually increased in the initial cycles. This is due to the incomplete dispersion of the electrolyte into the electrode materials at the beginning. At the 0.1 C rate, once the LiFePO_4 particles were fully saturated by the electrolyte, the capacity reached the maximum value of 146 mAh/g. After 100 cycles, the capacity of the coated LiFePO_4 electrode remained at 143.4 mAh/g, which is higher than that of the uncoated sample, 138 mAh/g. Furthermore, the capacity of the ZrO_2 coated LiFePO_4 has been significantly improved at the high charge/discharge rate. At the 1 C rate, the capacity of the coated electrode reached a maximum of 97 mAh/g and retained 94 mAh/g after 100 cycles, while the bare material only had a maximum capacity of 90 mAh/g and retained only 82 mAh/g after 100 cycles.

The mechanism behind the improvement of the electrochemical performance of the LiFePO_4 electrodes through

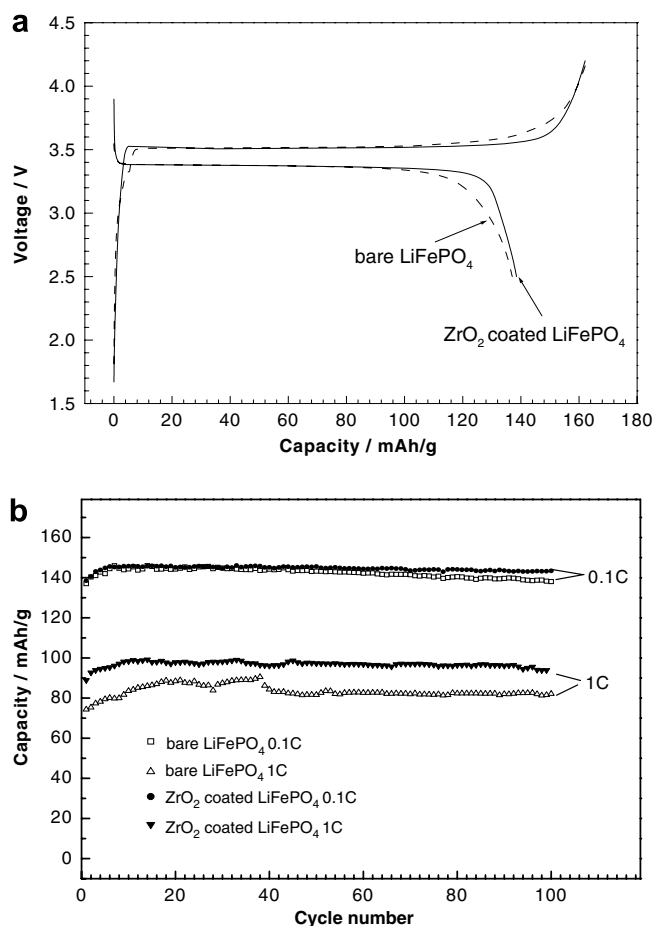


Fig. 4. The electrochemical performance of the bare and coated LiFePO₄ (a) first charge/discharge cycle at 0.1 C; (b) capacity retention after 100 cycles at 0.1 C and 1 C.

the ZrO₂ coating differs from that of carbon or polymer coatings. The ZrO₂ coating mitigates the volume change of the electrode during charge/discharge, which is similar to the effect of metal oxide coatings on other cathode materials [17–21]. When Li⁺ ions are extracted from LiFePO₄ electrode, the lattice contracts along the *a* and *b* axes, but expands along the *c* axis. The volume of the orthorhombic unit cell decreases by 6.64% when approaching the fully charged state (FePO₄[−]) [23]. The reverse process occurs during the discharging. Therefore, the LiFePO₄ crystals become fatigued over the repeated charge/discharge cycling, which induces decreased capacity. Since ZrO₂ has high mechanical toughness, the coated ZrO₂ layer can effectively prevent the fatigue of the core particles and sustain the structural stability on cycling. In particular, the ZrO₂ coating is very effective at withstanding the rapid strain changes within the LiFePO₄ cathode at high charge and discharge rates. A.c. electrochemical impedance spectroscopy (EIS) measurements indicated that the charge transfer resistance decreased remarkably after ZrO₂ coating. This is also evident from the flatter charge/discharge plateaux for the ZrO₂ coated LiFePO₄ electrode (Fig. 4a). The decrease in the interfacial resistance could also contribute to the enhancement of the high rate per-

mance. Furthermore, the ZrO₂ nanocoating is favorable to increasing the thermal stability by forming a more stable solid electrolyte interface (SEI) layer and covering the over-reactive sites on the particle surface to avoid probable electrolyte decomposition [17]. In addition, the ZrO₂ surface coating could also provide a protective layer for LiFePO₄ core particles to shield them from direct contact with the acidic electrolyte [24,25].

4. Conclusion

In summary, we prepared ZrO₂ coated LiFePO₄. HRTEM analysis confirmed that LiFePO₄ particles were coated with a layer of 2–3 nm ZrO₂. The ZrO₂ nanocoating increases the mechanical toughness of the core particles and decreases the interface charge transfer resistance, and thus, remarkably enhances the electrochemical performance of the LiFePO₄ cathode. Both the specific capacity and the cyclability of LiFePO₄ can be effectively improved, particularly at high rate. The nanocoating method could be a promising approach to improve the electrochemical properties of LiFePO₄ cathode materials.

Acknowledgment

This work was financially supported by the Australian Research Council (ARC) through an ARC Discovery project (DP0772999).

References

- [1] A.K. Padhi, K.S. Nanjundaswamy, J.B. Goodenough, *J. Electrochem. Soc.* 144 (1997) 1188.
- [2] A.K. Padhi, K.S. Nanjundaswamy, C. Masquelier, S. Okada, J.B. Goodenough, *J. Electrochem. Soc.* 144 (1997) 1609.
- [3] S.Y. Chung, J.T. Bloking, Y.M. Chiang, *Nat. Mater.* 1 (2002) 123.
- [4] J. Molenda, A. Stoklosa, T. Bak, *Solid State Ionics* 36 (1989) 53.
- [5] Y. Shimakawa, T. Numata, J. Tabuchi, *J. Solid State Chem.* 131 (1997) 138.
- [6] H. Huang, S.C. Yin, L.F. Nazar, *Electrochem. Solid State Lett.* 4 (2001) A170.
- [7] K. Zaghib, J. Shim, A. Guerfi, P. Charest, K.A. Striebel, *Electrochem. Solid State Lett.* 8 (2005) A207.
- [8] A.D. Spong, G. Vitins, J.R. Owen, *J. Electrochem. Soc.* 152 (2005) A2376.
- [9] H. Liu, P. Zhang, G.C. Li, Q. Wu, Y.P. Wu, *J. Solid State Electrochem.*, in press.
- [10] F. Croce, A.D. Epifanio, J. Hassoun, A. Deptula, T. Olczac, B. Scrosati, *Electrochem. Solid State Lett.* 5 (2002) A47.
- [11] G.X. Wang, L. Yang, Y. Chen, J.Z. Wang, S. Bewlay, H.K. Liu, *Electrochim. Acta* 50 (2005) 4649.
- [12] D.X. Gouveia, V. Lemos, J. A.C. de Paiva, A.G. Souza Filho, J. Mendes Filho, S.M. Lala, L.A. Montoro, J.M. Rosolen, *Phys. Rev. B* 72 (2005) 24105.
- [13] G.X. Wang, S. Bewlay, S.A. Needham, H.K. Liu, R.S. Liu, V.A. Drozd, J.F. Lee, J.M. Chen, *J. Electrochem. Soc.* 153 (2006) A25.
- [14] H. Liu, Q. Cao, L.J. Fu, C. Li, Y.P. Wu, H.Q. Wu, *Electrochem. Commun.* 8 (2006) 1553.
- [15] N. Ravet, A. Abouimrane, M. Armand, *Nat. Mater.* 2 (2003) 702.
- [16] M.S. Islam, D.J. Driscoll, C.A.J. Fisher, P.R. Slater, *Chem. Mater.* 17 (2005) 5085.
- [17] C. Li, H.P. Zhang, L.J. Fu, H. Liu, Y.P. Wu, E. Rahm, R. Holze, H.Q. Wu, *Electrochim. Acta* 51 (2006) 3872.

- [18] J. Cho, Y.J. Kim, T.J. Kim, B. Park, *Angew. Chem. Int. Ed.* 40 (2001) 3367.
- [19] J. Cho, T.J. Kim, Y.J. Kim, B. Park, *Electrochem. Solid State Lett.* 4 (2001) A159.
- [20] A.M. Kannan, A. Manthiram, *Electrochem. Solid State Lett.* 5 (2002) A167.
- [21] M.M. Thackeray, C.S. Johnson, J.S. Kim, *Electrochem. Commun.* 5 (2003) 752.
- [22] N. Ohta, K. Takada, L.Q. Zhang, R.Z. Ma, M. Osada, T. Sasaki, *Adv. Mater.* 18 (2006) 2226.
- [23] A.S. Anderson, B. Kalska, L. Häggström, J.O. Thomas, *Solid State Ionics* 130 (2000) 41.
- [24] Z.X. Wang, L.J. Liu, L.Q. Chen, X.J. Huang, *Solid State Ionics* 148 (2002) 335.
- [25] Z.X. Wang, X.J. Huang, L.Q. Chen, *J. Electrochem. Soc.* 150 (2003) A199.